

Name: _____ Teacher: _____

YEAR 11 MATHEMATICS EXTENSION 1 ASSESSMENT TASK – JUNE 2009

Time Allowed: 70 minutes

Full working should be shown in every question.

Marks may be deducted for careless or badly arranged work.

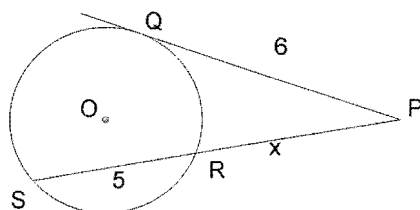
No liquid paper is to be used.

If a correction is to be made, one line is to be ruled through the incorrect answer.

Marks

1. PQ is a tangent to the circle below. Find x giving reasons where necessary.

3



2. Find the acute angle between the lines $x - 2y + 1 = 0$ and $x + 3y + 2 = 0$.

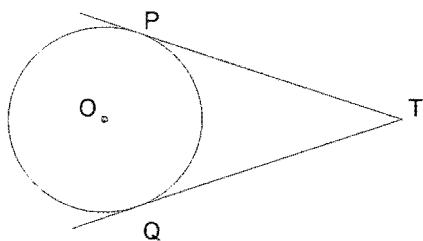
3

3. Solve $2\tan\theta - 3\cot\theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$.

3

4. PT and QT are tangents to a circle with centre O. Prove $\angle PTQ = 2\angle PQO$

3



5. Express $2\cos\theta + 2\sqrt{3}\sin\theta$ in the form $R\cos(\theta - \alpha)$ and hence find the minimum value of the expression.

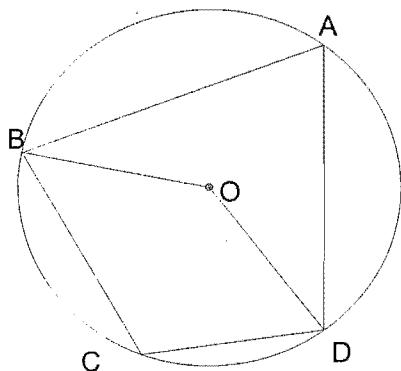
3

6. Prove, using $t = \tan\frac{\theta}{2}$, that $\frac{1 + \operatorname{cosec}\theta}{\cot\theta} = \frac{1 + \tan\frac{\theta}{2}}{1 - \tan\frac{\theta}{2}}$

3

7. The point $P(-2,9)$ divides the interval AB in the ratio $k:1$, where A is $(7,-3)$ and B is $(1,5)$. Find the value of k . 2

8. $\angle BAD = \alpha$ and $\angle BOD = \angle BCD$. Find α giving reasons. 3



9. Prove $\frac{\sin A}{\cos A + \sin A} + \frac{\sin A}{\cos A - \sin A} = \tan 2A$ 3

10. The angle between two lines $y = mx$ and $y = \frac{1}{2}x$ is 45° . Find the possible values of m . 3

11. Solve $\sqrt{2}\sin\theta - \cos\theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$ 3

12. AB and CD are two intersecting chords of a circle and CD is parallel to the tangent to the circle at B .

(a) Draw a neat sketch of the above information.

- (b) Prove that AB bisects $\angle CAD$. 3

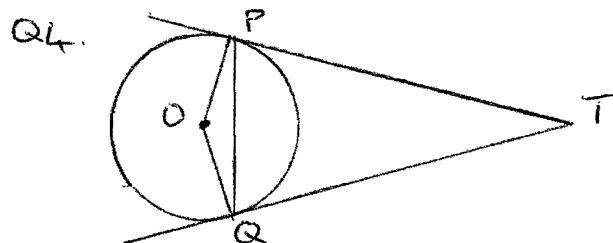
END OF TEST

Q1. $x(x+9) = 6^2$ (sq. of tangent
 $x^2 + 9x - 36 = 0$ = product of intercepts)
 $(x+9)(x-4) = 0$
 $x = -9$ or 4
 since $x > 0$, $x = 4$ (3)

Q2 $x - 2y + 1 = 0 \therefore y = \frac{1}{2}x + \frac{1}{2} \therefore m_1 = \frac{1}{2}$
 $x + 3y + 2 = 0 \therefore y = -\frac{1}{3}x - \frac{2}{3} \therefore m_2 = -\frac{1}{3}$
 $\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{\frac{1}{2} - (-\frac{1}{3})}{1 + \frac{1}{2} \times (-\frac{1}{3})} \right|$
 $= \left| \frac{\frac{5}{6}}{\frac{5}{6}} \right|$
 $= 1$ (3)

$\therefore \alpha = 45^\circ$
 Q3 $2 \tan \theta - \frac{3}{\tan \theta} = 5$
 $2 \tan^2 \theta - 3 = 5 \tan \theta$
 $2 \tan^2 \theta - 5 \tan \theta - 3 = 0$
 $2 \tan^2 \theta - 6 \tan \theta + \tan \theta - 3 = 0$
 $2 \tan \theta (\tan \theta - 3) + 1(\tan \theta - 3) = 0$
 $(2 \tan \theta + 1)(\tan \theta - 3) = 0$
 $\therefore \tan \theta = -\frac{1}{2}$ or $\tan \theta = 3$ (3)

$\therefore \theta = 153^\circ 26', 333^\circ 26', 71^\circ 34', 251^\circ 34'$
 In order:
 $\theta = 71^\circ 34', 153^\circ 26', 251^\circ 34', 333^\circ 26'$



Let $\angle PQO = \alpha$
 $\therefore \angle PQT = 90^\circ - \alpha$ ($\angle$ bet. tang & radius = 90°)
 ΔPQT is isosceles (tang. from ext. pt. are equal)
 $\therefore \angle QPT = 90^\circ - \alpha$ (eq. $\angle$'s of isos. Δ)
 $\therefore \angle PTQ = 180^\circ - 2(90^\circ - \alpha)$ ($\angle$ sum of Δ)
 $= 2\alpha$
 $\therefore \angle PTQ = 2 \times \angle PQO$ (3)

Q5. $2 \cos \theta + 2\sqrt{3} \sin \theta = R \cos(\theta - \alpha)$
 $= R \cos \theta \cos \alpha + R \sin \theta \sin \alpha$
 $\therefore R \cos \alpha = 2, R \sin \alpha = 2\sqrt{3}$
 $\therefore R = \sqrt{2^2 + (2\sqrt{3})^2} = 4$
 and $\tan \alpha = \frac{2\sqrt{3}}{2} = \sqrt{3} \therefore \alpha = 60^\circ$
 $\therefore \text{Exp.} = 4 \cos(\theta - 60^\circ)$
 Min. Value = -4 (3)

Q6. If $\tan \frac{\theta}{2} = \frac{t}{1}$
 then $\cos \theta = \frac{1}{\sin \theta}$
 $= \frac{1}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$
 $= \frac{1}{2 \cdot \frac{t}{\sqrt{1+t^2}} \cdot \frac{1}{\sqrt{1+t^2}}}$
 $= \frac{1+t^2}{2t}$
 and $\cot \theta = \frac{1}{\tan \theta}$
 $= \frac{1}{\frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}}$
 $= \frac{1-t^2}{2t}$
 $\therefore \text{LHS} = 1 + \frac{1+t^2}{\frac{1-t^2}{2t}}$
 $= \frac{2t + 1+t^2}{1-t^2}$
 $= \frac{(1+t)^2}{(1+t)(1-t)}$
 $= \frac{1+t}{1-t}$
 $= \text{RHS}$ (3)

Q7. Using x-values and $m=k, n=1$
 $x_p = \frac{mx_2 + nx_1}{m+n}$
 $-2 = \frac{k \times 1 + 1 \times 7}{k+1}$
 $-2k - 2 = k + 7$
 $-3k = 9$
 $k = -3$ (2)

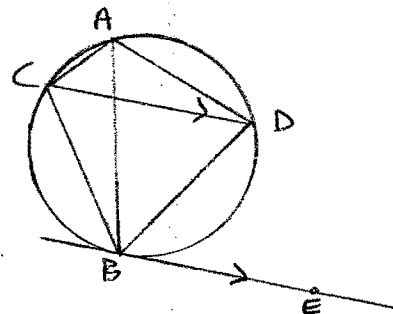
Q8. $\angle BAD = x$ ✓
 $\therefore \angle BOD = 2x$ ($\angle$ at centre = $2 \times \angle$ at circum. stand. on same arc)
 $\angle BCD = 180^\circ - x$ (opp. $\angle$'s of a cyclic quad are suppl.)
 But $\angle BOD = \angle BCD$ (given)
 $\therefore 2x = 180^\circ - x$
 $3x = 180^\circ$
 $x = 60^\circ$ — (3)

Q9. $LHS = \frac{\sin A \cos A - \sin^2 A + \cos A \sin A + \sin^2 A}{\cos^2 A - \sin^2 A}$ ✓
 $= \frac{2 \sin A \cos A}{\cos^2 A - \sin^2 A}$
 $= \frac{\sin 2A}{\cos 2A}$ —
 $= \tan 2A$
 $= RHS$ (3)

Q10. $\left| \frac{m - \frac{1}{2}}{1 + m + \frac{1}{2}} \right| = \tan 45^\circ = 1$ —
 $\frac{m - \frac{1}{2}}{1 + \frac{1}{2}m} = \pm 1$
 $\therefore m - \frac{1}{2} = 1 + \frac{1}{2}m$ or $m - \frac{1}{2} = -1 - \frac{1}{2}m$
 $\frac{1}{2}m = \frac{3}{2}$ or $\frac{3}{2}m = -\frac{1}{2}$ —
 $m = 3$ or $m = -\frac{1}{3}$ (3)

Q11. let $\sqrt{2} \sin \theta - \cos \theta = R \sin(\theta - \alpha)$
 $= R \sin \theta \cos \alpha - R \cos \theta \sin \alpha$
 $\therefore R \cos \alpha = \sqrt{2}$, $R \sin \alpha = 1$
 $\therefore R = \sqrt{2+1} = \sqrt{3}$
 and $\tan \alpha = \frac{1}{\sqrt{2}} \therefore \alpha = 35^\circ 16'$ —
 $\therefore \sqrt{3} \sin(\theta - 35^\circ 16') = 1$
 $\sin(\theta - 35^\circ 16') = \frac{1}{\sqrt{3}}$ —
 $\therefore \theta - 35^\circ 16' = 35^\circ 16', 144^\circ 44'$
 $\therefore \theta = 70^\circ 32', 180^\circ$ — (3)

Q12. (a)



(b) Draw AC, AD, BC, BD
 let $\angle DBE = x$ ✓
 $\therefore \angle BAD = x$ ($\angle$ bet. tang & chord = $\angle$ in altern. seg.)
 $\angle BDC = x$ (Altern. $\angle$'s, $CD \parallel BE$)
 $\angle BAC = x$ ($\angle$'s on circum stand on the same arc BC)
 $\therefore \angle BAD = \angle BAC$ (3)

Alternative Q11 soln. ✓

If $\sqrt{2} \sin \theta - \cos \theta = 1$
 let $t = \tan \frac{\theta}{2}$
 $\therefore \sin \theta = \frac{2t}{1+t^2}$, $\cos \theta = \frac{1-t^2}{1+t^2}$
 $\therefore \sqrt{2} \times \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} = 1$ —
 $2\sqrt{2}t - 1 + t^2 = 1 + t^2$
 $2\sqrt{2}t = 2$
 $\tan \frac{\theta}{2} = \frac{1}{\sqrt{2}}$
 $\therefore \frac{\theta}{2} = 35^\circ 16'$
 $\theta = 70^\circ 32'$ —

Check $\theta = 180^\circ$ —
 $LHS = \sqrt{2} \times 0 - 1 = -1 = RHS$
 $\therefore \theta = 70^\circ 32', 180^\circ$ (3)